

你要分析：

分析	做法
Late vs on-time review score	比较平均分、中位数、分布
Bad review rate	主指标： <code>review_score <= 2</code>
Low review rate	sensitivity: <code>review_score <= 3</code>
Delay severity vs review score	看延迟越严重，评分是否越低
Delay severity vs bad review rate	看严重延迟是否对应更高差评率

文档明确建议：主 bad review metric 用 1–2 星，1–3 星作为 sensitivity metric。
统计检验不要堆太多。够用就行：

```
Mann-Whitney U: late vs on-time review score
Chi-square: late flag vs bad review flag
Kruskal-Wallis: delay severity bands vs review score
Optional logit: bad review ~ is_late + controls
```

文档也提醒：不要为了显得复杂而过度使用统计测试，只用能支持核心 business claim 的测试。

Figure 1: image.png

Zeash + Moses Gap Matrix

Zeash: Strong Moses: Strong Existing projects show relationship but stop at analytical conclusion Your contribution: Use as core CX harm evidence and trigger for recovery design

```
rg -n "review_score|bad_review|low_review|rating|satisfaction|late.review/delay.review|Mann|Whitney|Wilcoxon|chi-square|chisq|Kruskal|OLS|logit|logistic" references/source_projects
```

1. late vs on-time review score
2. bad review
3. 1–2 1–3
- 4.
5. claim association causation

bad_review = review_score <= 2

low_review = review_score <= 3 Gap Matrix Zeash 1–2 stars Moses 1–3 stars
main bad review = 1–2 optional sensitivity = 1–3

late delivery is associated with lower review score

This relationship is used as evidence of customer harm and becomes the trigger logic for service recovery design.

review score	recovery trigger
--------------	------------------

03 Review Impact Analysis

Business Objective

This notebook evaluates whether delivery failure is associated with customer dissatisfaction.

The purpose is not to prove causality.

The Olist dataset is observational, therefore this analysis measures association between delivery delay and customer experience outcomes.

The business question is:

Should late delivery be treated as a customer experience early-warning signal?

This notebook extends previous delivery diagnosis by translating operational failure into customer harm evidence.

The analysis compares:

- Late delivery orders
- On-time / early delivery orders

using:

1. Review score differences
2. Bad review risk differences
3. Delay severity patterns
4. Statistical association tests

The findings will support the later service recovery design.

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt

from scipy.stats import mannwhitneyu
```

```
from scipy.stats import chi2_contingency
from scipy.stats import kruskal
```

1. Analysis Population

The analysis uses delivered orders with:

- valid delivery timestamps;
- valid estimated delivery date;
- valid review score.

The population follows the delivery analysis definition established in previous notebooks.

Late delivery is treated using calendar dates rather than raw timestamps to avoid false late classification.

```
df = pd.read_csv(
    "../data/processed/master_order_table.csv"
)

df.shape
```

```
(99441, 43)
```

```
df.columns.tolist()
```

```
['order_id',
 'customer_id',
 'order_status',
 'order_purchase_timestamp',
 'order_approved_at',
 'order_delivered_carrier_date',
 'order_delivered_customer_date',
 'order_estimated_delivery_date',
 'customer_unique_id',
 'customer_zip_code_prefix',
 'customer_city',
 'customer_state',
 'item_count',
 'product_count',
```

```
'seller_count',
'category_count',
'total_item_price',
'total_freight_value',
'is_multi_seller',
'is_multi_category',
'payment_record_count',
'payment_type_count',
'total_payment_value',
'max_payment_installments',
'payment_types',
'review_id',
'review_score',
'review_comment_title',
'review_comment_message',
'review_creation_date',
'review_answer_timestamp',
'review_record_count',
'has_multiple_reviews',
'has_item_record',
'has_payment_record',
'has_review_record',
'actual_delivery_date',
'estimated_delivery_date',
'delivery_analysis_eligible',
'delay_days',
'is_late',
'is_late_timestamp_rule',
'delay_band']
```

```
analysis_df = df[
    (df["order_status"]=="delivered")
    &
    (df["review_score"].notna())
    &
    (df["delay_days"].notna())
].copy()
```

```
analysis_df.shape
```

```
(95824, 43)
```

Delivery Status

Late delivery: `delay_days > 0`

On-time / Early: `delay_days <= 0`

Customer Dissatisfaction Signals

Primary metric: `bad_review = review_score <= 2` Reason: 1–2 star reviews represent stronger dissatisfaction. Sensitivity metric: `low_review = review_score <= 3` Reason: This follows alternative satisfaction grouping used in previous reference analysis.

```
analysis_df["delivery_status"] = np.where(
    analysis_df["delay_days"] > 0,
    "Late",
    "On-time/Early"
)

analysis_df["bad_review"] = (
    analysis_df["review_score"] <= 2
)

analysis_df["low_review"] = (
    analysis_df["review_score"] <= 3
)

analysis_df.head()
```

	order_id	customer_id	order_status	order_purchase
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered	2017-10-02 10
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	2018-07-24 20
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	2018-08-08 08
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	2017-11-18 19
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered	2018-02-13 21

3. Descriptive CX Impact

Review Score Comparison

Question:

Do late orders receive lower customer ratings?

Average review score provides the first indication of customer experience difference.

However, average scores alone do not measure dissatisfaction risk, therefore additional harm signals are analyzed later.

```
review_summary = (
    analysis_df
    .groupby("delivery_status")
    ["review_score"]
    .agg(
        orders="count",
        average_review_score="mean",
        median_review_score="median"
    )
    .reset_index()
)
```

review_summary

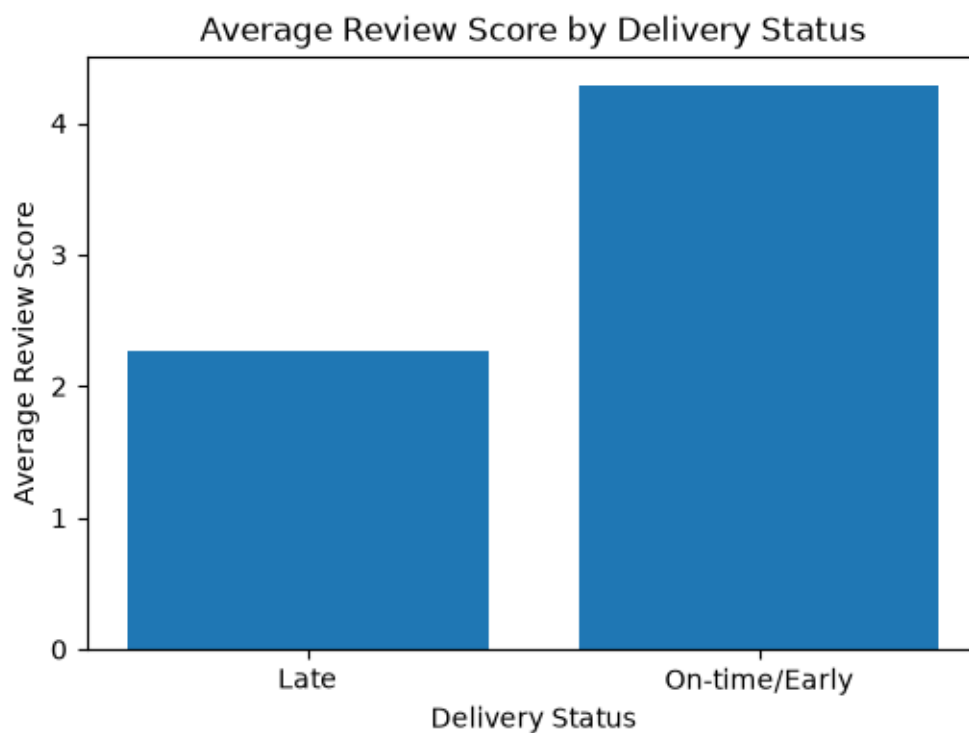
	delivery_status	orders	average_review_score	median_review_score
0	Late	6381	2.270491	1.0
1	On-time/Early	89443	4.290386	5.0

```
plt.figure(figsize=(6,4))

plt.bar(
    review_summary["delivery_status"],
    review_summary["average_review_score"]
)

plt.ylabel(
    "Average Review Score"
```

```
)  
  
plt.xlabel(  
    "Delivery Status"  
)  
  
plt.title(  
    "Average Review Score by Delivery Status"  
)  
  
plt.show()
```



Interpretation

If late orders receive lower review scores, this suggests delivery reliability is associated with weaker customer experience.

However, review score difference alone does not show the magnitude of customer harm.

Therefore, dissatisfaction risk metrics are analyzed next.

4. Customer Harm Signal Analysis

Bad Review Risk

Customer dissatisfaction is operationalized as: Bad Review = review_score ≤ 2

The analysis measures: 1. Bad review rate 2. Risk ratio 3. Risk difference

These metrics translate review impact into business-relevant customer harm signals.

```
bad_review_summary = (
    analysis_df
    .groupby("delivery_status")
    .agg(
        total_reviews=("review_score", "count"),
        bad_reviews=("bad_review", "sum")
    )
    .reset_index()
)

bad_review_summary["bad_review_rate"] = (
    bad_review_summary["bad_reviews"]
    /
    bad_review_summary["total_reviews"]
)

bad_review_summary
```

	delivery_status	total_reviews	bad_reviews	bad_review_rate
0	Late	6381	3983	0.624197
1	On-time/Early	89443	8289	0.092674

```
late_rate = (
    bad_review_summary
    .query("delivery_status=='Late'")
    ["bad_review_rate"]
    .iloc[0]
)
```

```

ontime_rate = (
    bad_review_summary
    .query("delivery_status=='On-time/Early'")
    ["bad_review_rate"]
    .iloc[0]
)

risk_ratio = late_rate / ontime_rate

risk_difference = late_rate - ontime_rate

print(
    "Risk Ratio:",
    round(risk_ratio,2)
)

print(
    "Risk Difference:",
    round(risk_difference*100,2),
    "percentage points"
)

```

Risk Ratio: 6.74

Risk Difference: 53.15 percentage points

Sensitivity Analysis: Alternative Dissatisfaction Definition

Reference projects use different dissatisfaction thresholds:

- Zeash project:
 - bad review = review_score <= 2
- Moses project:
 - low review = review_score <= 3

Therefore, this project uses:

Primary metric:

```
bad_review = review_score <= 2
```

Sensitivity metric:

```
low_review = review_score <= 3
```

The purpose is to test whether the relationship between late delivery and dissatisfaction remains under a broader definition.

```
sensitivity_summary = (  
    analysis_df  
    .groupby("delivery_status")  
    .agg(  
        total_orders=("review_score", "count"),  
        bad_review_rate=("bad_review", "mean"),  
        low_review_rate=("low_review", "mean")  
    )  
    .reset_index()  
)
```

```
sensitivity_summary
```

	delivery_status	total_orders	bad_review_rate	low_review_rate
0	Late	6381	0.624197	0.732957
1	On-time/Early	89443	0.092674	0.173418

```
late_low_rate = (  
    sensitivity_summary  
    .query("delivery_status=='Late'")  
    ["low_review_rate"]  
    .iloc[0]  
)
```

```
ontime_low_rate = (  
    sensitivity_summary  
    .query("delivery_status=='On-time/Early'")  
    ["low_review_rate"]
```

```

        .iloc[0]
    )

    low_review_risk_ratio = (
        late_low_rate /
        ontime_low_rate
    )

    print(
        "Low review risk ratio (<=3 stars):",
        round(low_review_risk_ratio, 2)
    )

```

Low review risk ratio (<=3 stars): 4.23

Interpretation

If both definitions show higher dissatisfaction risk among late deliveries:

- strong dissatisfaction (1–2 stars)
- broader dissatisfaction (1–3 stars)

then the relationship is less likely to depend on one arbitrary threshold.

This strengthens the interpretation that delivery reliability is associated with customer experience outcomes.

6. Mann-Whitney U Test

Question:

Do late and on-time orders have different review score distributions?

Why Mann-Whitney:

Review score is an ordinal 1–5 rating scale.

The test compares whether the distributions of review scores differ between two independent delivery groups.

Null hypothesis:

Review score distributions are the same between late and on-time orders.

Alternative hypothesis:

Review score distributions differ between delivery groups.

```
late_scores = (
    analysis_df
    .query("delivery_status=='Late'")
    ["review_score"]
)

ontime_scores = (
    analysis_df
    .query("delivery_status=='On-time/Early'")
    ["review_score"]
)

mw_stat, mw_p = mannwhitneyu(
    late_scores,
    ontime_scores,
    alternative="two-sided"
)

print(
    "Mann-Whitney statistic:",
    mw_stat
)

print(
    "p-value:",
    mw_p
)
```

Mann-Whitney statistic: 103305546.0

p-value: 0.0

```
if mw_p < 0.05:
    print(
        "Review score distributions differ significantly between delivery groups."
```

```
)  
else:  
    print(  
        "No statistically significant difference detected."  
    )
```

Review score distributions differ significantly between delivery groups.

7. Chi-square Test

Question:

Is late delivery associated with a higher probability of bad reviews?

Variables:

Delivery status:

- Late
- On-time/Early

Customer outcome:

- Bad review
- Not bad review

Null hypothesis:

Delivery status and bad review occurrence are independent.

Alternative hypothesis:

Delivery status and bad review occurrence are associated.

```
contingency = pd.crosstab(  
    analysis_df["delivery_status"],  
    analysis_df["bad_review"]  
)
```

```
contingency
```

bad_review	False	True
delivery_status		
Late	2398	3983
On-time/Early	81154	8289

```
chi2, chi_p, dof, expected = chi2_contingency(
    contingency
)

print(
    "Chi-square statistic:",
    chi2
)

print(
    "p-value:",
    chi_p
)
```

Chi-square statistic: 15064.148966139202
p-value: 0.0

```
if chi_p < 0.05:
    print(
        "Bad review occurrence is significantly associated with delivery status."
    )
else:
    print(
        "No significant association detected."
    )
```

Bad review occurrence is significantly associated with delivery status.

Optional Robustness Check

Logistic Regression

This analysis is included as a robustness check.

The main evidence in this notebook comes from:

- Review score comparison
- Bad review risk analysis
- Mann-Whitney U test
- Chi-square test
- Delay severity analysis

The logistic regression provides additional validation by examining whether late delivery remains associated with bad review probability after controlling for observable order characteristics.

The model estimates:

`bad_review ~ is_late + operational controls`

Control variables include:

- total item price
- freight value
- item quantity
- seller complexity
- category complexity

Important limitation:

This model does not prove causal impact.

Because the Olist dataset is observational, results should be interpreted as association:

“Late delivery remains associated with higher bad review probability after accounting for observable factors.”

Unobserved factors such as customer expectations, communication quality, and logistics events are not included.

```
import statsmodels.formula.api as smf
import numpy as np

model_df = analysis_df[
    [
        "bad_review",
        "is_late",
        "total_item_price",
        "total_freight_value",
        "item_count",
        "seller_count",
```

```

        "category_count"
    ]
].dropna().copy()

model_df["bad_review"] = (
    model_df["bad_review"]
    .astype(int)
)

model_df["is_late"] = (
    model_df["is_late"]
    .astype(int)
)

model_df.dtypes

```

```

bad_review          int64
is_late             int64
total_item_price    float64
total_freight_value float64
item_count          float64
seller_count        float64
category_count      float64
dtype: object

```

```

logit_model = smf.logit(
    """
    bad_review ~
    is_late
    + total_item_price
    + total_freight_value
    + item_count
    + seller_count
    + category_count
    """,
    data=model_df
).fit()

```

```
logit_model.summary()
```

Optimization terminated successfully.
 Current function value: 0.321399
 Iterations 6

Dep. Variable:	bad_review	No. Observations:	95824
Model:	Logit	Df Residuals:	95817
Method:	MLE	Df Model:	6
Date:	Sun, 27 Sep 2026	Pseudo R-squ.:	0.1602
Time:	10:12:48	Log-Likelihood:	-30798.
converged:	True	LL-Null:	-36672.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-4.5163	0.082	-55.111	0.000	-4.677	-4.356
is_late	2.8851	0.029	100.475	0.000	2.829	2.941
total_item_price	0.0002	5.1e-05	3.028	0.002	5.45e-05	0.000
total_freight_value	0.0020	0.001	3.757	0.000	0.001	0.003
item_count	0.4303	0.018	23.324	0.000	0.394	0.466
seller_count	1.4798	0.076	19.582	0.000	1.332	1.628
category_count	0.1146	0.100	1.149	0.251	-0.081	0.310

```
odds_ratio = np.exp(
    logit_model.params
)

odds_ratio.sort_values(
    ascending=False
)
```

```
is_late          17.906122
seller_count     4.391891
item_count       1.537777
category_count   1.121429
total_freight_value 1.002027
total_item_price 1.000154
Intercept        0.010930
dtype: float64
```

Interpretation

The logistic regression provides robustness evidence.

The odds ratio of `is_late` shows whether late delivery remains associated with bad review probability after controlling for observable order characteristics.

An odds ratio above 1 indicates higher odds of receiving a bad review among late delivery orders, holding other included variables constant.

This strengthens the customer harm evidence used for recovery trigger design.

However, the model does not establish causal impact because other unobserved customer experience factors may influence review outcomes.

8. Delay Severity and Customer Experience

Late delivery is not a binary problem.

A one-day delay and a thirty-day delay may create different levels of customer dissatisfaction.

Therefore, this analysis evaluates whether increasing delay severity corresponds to worsening review outcomes.

```
bins=[
    -999,
    0,
    3,
    7,
    14,
    30,
    999
]

labels=[
    "On-time/Early",
    "1-3 days late",
    "4-7 days late",
    "8-14 days late",
    "15-30 days late",
    "30+ days late"
]
```

```
analysis_df["delay_band"] = pd.cut(
    analysis_df["delay_days"],
    bins=bins,
    labels=labels
)

analysis_df["delay_band"].value_counts()
```

```
delay_band
On-time/Early      89443
1-3 days late      1852
4-7 days late      1748
8-14 days late     1446
15-30 days late    1006
30+ days late       329
Name: count, dtype: int64
```

```
severity_summary = (
    analysis_df
    .groupby(
        "delay_band",
        observed=True
    )
    .agg(
        orders=("review_score", "count"),
        average_review_score=("review_score", "mean"),
        bad_review_rate=("bad_review", "mean")
    )
    .reset_index()
)

severity_summary
```

	delay_band	orders	average_review_score	bad_review_rate
0	On-time/Early	89443	4.290386	0.092674
1	1-3 days late	1852	3.291037	0.321274
2	4-7 days late	1748	2.102975	0.676773
3	8-14 days late	1446	1.670816	0.801521

	delay_band	orders	average_review_score	bad_review_rate
4	15-30 days late	1006	1.614314	0.818091
5	30+ days late	329	2.057751	0.677812

```
plt.figure(figsize=(10,5))

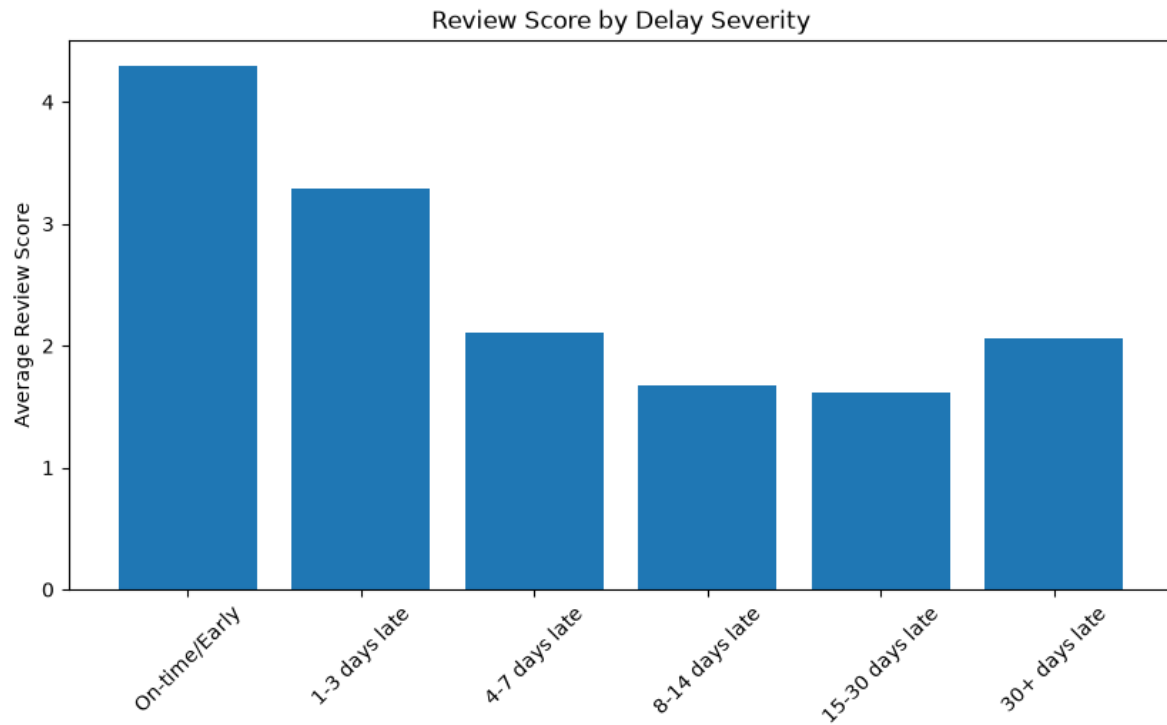
plt.bar(
    severity_summary["delay_band"].astype(str),
    severity_summary["average_review_score"]
)

plt.xticks(
    rotation=45
)

plt.ylabel(
    "Average Review Score"
)

plt.title(
    "Review Score by Delay Severity"
)

plt.show()
```



```
severity_groups = [
    group["review_score"].values
    for name, group
    in analysis_df.groupby(
        "delay_band",
        observed=True
    )
]

kruskal_stat, kruskal_p = kruskal(
    *severity_groups
)

print(
    "Kruskal-Wallis statistic:",
    kruskal_stat
)
```

```
print(  
    "p-value:",  
    kruskal_p  
)
```

```
Kruskal-Wallis statistic: 10119.654793870715  
p-value: 0.0
```

Interpretation

A significant Kruskal-Wallis result indicates that review score distributions differ across delay severity groups.

The pattern is important for service recovery design:

Small delays may require communication and expectation management.

Severe delays may require escalation, compensation consideration, or priority intervention.

Therefore, delay severity can be used as a recovery prioritization signal.

9. Recovery Trigger Translation

The purpose of this analysis is not only to identify statistical relationships.

The findings are translated into operational recovery logic.

Evidence chain:

Delivery failure

↓

Customer harm signal

↓

Recovery trigger

↓

Operational action

Recovery Trigger Framework

Evidence Signal	Customer Risk Interpretation	Recovery Action
Late delivery detected	Delivery promise failure	proactive delay notification
Higher bad review probability	Increased dissatisfaction risk	recovery outreach
Severe delay band	Escalating customer harm	priority escalation
High-risk seller pattern	Repeated operational failure	seller governance intervention
High-risk category/region	Structural delivery weakness	targeted improvement plan

This converts analytical findings into a service recovery operating model.

10. CX Impact Summary

Evidence 1: Review Score Impact

Late delivery orders received a lower average review score (2.27) compared with on-time/early orders (4.29).

Evidence 2: Customer Harm Signal

Late orders showed a bad review rate of 62.4%, compared with 9.3% for on-time/early orders.

This represents a 6.74x higher bad review risk and a 53.15 percentage point difference.

Evidence 3: Robustness

The logistic regression shows that late delivery remains associated with higher bad review odds after controlling for order characteristics.

The estimated odds ratio for is_late is approximately 17.9.

Evidence 4: Severity Pattern

Kruskal-Wallis testing indicates that review score distributions differ across delay severity groups.

This supports differentiated recovery actions based on delay severity.

Business Implication

Delivery failure should not only be monitored as an operational KPI.

It can be treated as an early-warning signal for customer dissatisfaction and a trigger for proactive service recovery.

Limitation

This analysis identifies association rather than causal impact.

Other factors such as seller performance, product category, logistics conditions, and customer expectations may also influence review outcomes.

The recovery system proposed later requires future validation through operational testing.